

Unshrouding: Evidence from Bank Overdrafts in Turkey

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ABSTRACT

Lower prices produce higher demand . . . or do they? A bank's direct marketing to holders of "free" checking accounts shows that a large discount on 60% APR overdrafts *reduces* overdraft usage, especially when bundled with a discount on debit card or autodebit transactions. In contrast, messages mentioning overdraft availability without mentioning price *increase* usage. Neither change persists long after the messages stop. These results do not square easily with classical models of consumer choice and firm competition. Instead, they support behavioral models where consumers underestimate and are inattentive to overdraft costs, and firms respond by shrouding overdraft prices in equilibrium.

MANY BUSINESS MODELS IN FINANCIAL services and other industries rely on expensive add-ons that are tied to base goods. Examples include expensive overdraft credit (add-on) tied to a "free" checking account (base good), back-end management fees tied to "free" investment advice, printer cartridges and printers, luggage fees and airline tickets, and dealer-supplied maintenance and automobiles. A closely related practice is overage/penalty pricing. Businesses with such revenue models typically focus their marketing and competitive strategy on the base good, even though add-ons/overages can be critical revenue sources.

Retail banking provides a striking example: banks often market checking accounts as free, even though many consumers end up paying high fees for overdraft credit.¹ Government audits find that banks rarely market overdraft services at the customer acquisition stage and even actively discourage employees from providing information on overdraft terms (General Accounting

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¹ An overdraft occurs if the checking account holder initiates a transaction that makes her balance negative or more negative.

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Office (2008), Competition and Markets Authority (2014), *Consumer Financial Protection Bureau v. TCF National Bank* (2017)). After acquiring customers, banks in some markets including Turkey, the site of our experiment, blur the line between positive and negative balances by highlighting for customers an available-to-withdraw figure that adds the available overdraft credit line amount to the checking account balance, while making information on disaggregated balances and finance charges more difficult to find. Turkish banks during our study period basically drew attention to overdrafts only when promoting the feature to existing customers. Yet, even those promotions did not explicitly mention the price of overdraft credit.

Why would a bank hide information on overdraft costs? After all, a classically rational consumer would simply infer that shrouded prices are high prices. But recent behavioral theories show that shrouded and high prices can persist if consumers tend to underestimate their add-on costs and firms cannot profit from debiasing consumers with more transparent pricing or information about competitors' high add-on prices (Gabaix and Laibson (2006), Grubb (2015), Heidhues, Köszegi, and Murooka (2017)).²

Bank regulators respond to such practices in various ways. For example, in the United States regulators require upfront consumer opt-in for debit card and ATM overdrafts, in both the United States and United Kingdom regulators caution banks against relying too heavily on overdraft revenue, and in Turkey regulators cap overdraft prices.

But empirical evidence is lacking on key questions raised by theory and policy. Do consumers actually underestimate costs of add-ons such as overdrafts?, do firms actually have incentives to shroud add-on prices instead of competing to debias consumers?, how do consumers allocate attention to add-ons?, and how quickly does consumer learning about add-ons break a shrouded equilibrium? In short, empirical evidence on what drives overdraft pricing, advertising, and usage is limited and largely descriptive.

These questions are central to overdraft markets. Beginning in the 1990s, overdraft revenue replaced monthly subscription fees as banks' major source of explicit income from checking accounts, shifting the pricing equilibrium for retail banking in much of the world to "free if nonnegative balance, very expensive if in overdraft." In the United States, banks collect more than \$10 billion in overdraft revenue annually. In the United Kingdom, banks derive almost as much income from overdrafts as from reinvesting checking account deposits (Competition and Markets Authority (2014)). In Turkey, the site of our experiment, the announcement of a price ceiling on overdrafts was immediately followed by a 1.4% reduction in bank share prices, with a 2.1% drop for the most overdraft-reliant bank (Toksabay (2013)).

We worked with Yapi Kredi (YK), one of the largest banks in Turkey, to design a randomized direct marketing experiment that distinguishes between

² For related evidence on consumer perceptions of overdraft costs, see Armstrong and Vickers (2012) and Stango and Zinman (2014). For related models of limited and reactive consumer attention, see, for example, Gabaix (2014) and Bordalo, Gennaioli, and Shleifer (2015).

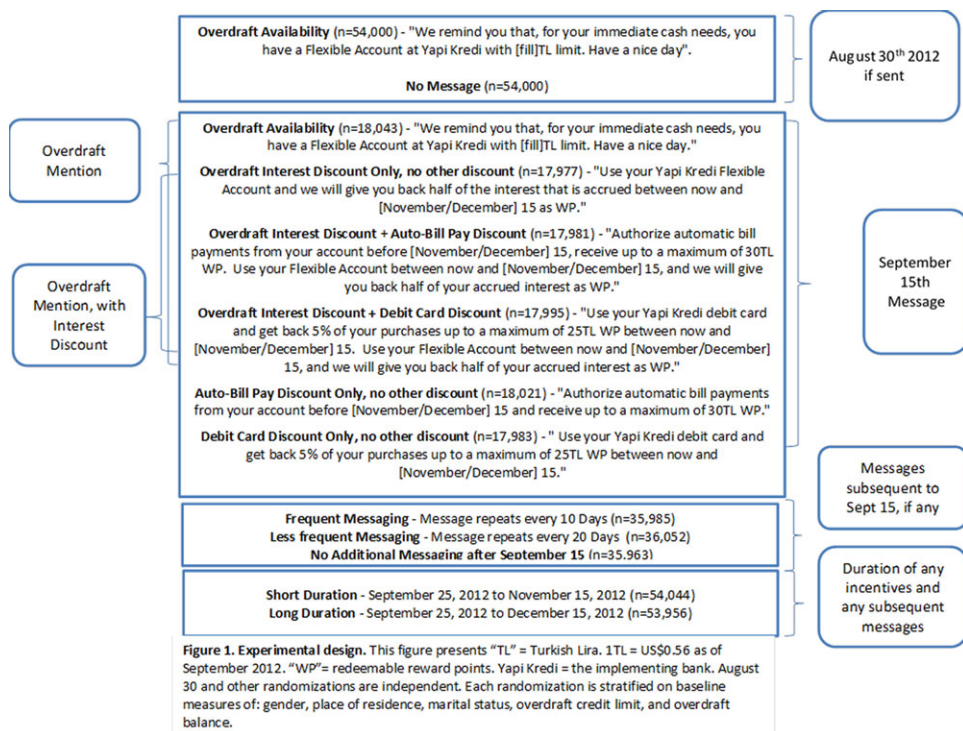


Figure 1. Experimental design. (Color figure can be viewed at wileyonlinelibrary.com)

classical and behavioral models of add-on pricing and advertising. YK sought to learn more about its optimal strategy for pricing and advertising its overdraft product. The bank was particularly interested in understanding whether its past promotional pricing and advertising content tactics were effective in increasing demand, and if not why not. YK's interest rate (60% APR) and product design was in line with standard practices and regulations. As is common in overdraft markets, the product was priced expensively relative to seemingly close substitutes (like credit cards), and disproportionally to credit risk (as recently found by Turkish authorities).

YK sought to target marginal overdrawers among its existing client base, and hence the experiment varied the promotions that YK sent via SMS (text message) from September to December 2012 to 108,000 existing checking account clients who had not overdrawn during the previous few months. These clients are likely representative of a substantial population of marginal overdrawers in Turkey, and they share key characteristics with "banked" populations in both more- and less-developed countries.

The experimental design produces random variation, both across clients and over time, in overdraft prices and in messaging content, frequency, and duration (see Figure 1). Our tests rely on overdraft usage comparisons across groups receiving different promotions, since Turkish banks frequently use

SMS-based promotions (a pure control group would be off-equilibrium). The key comparison is between overdraft-promoting messages that mention price and those that do not.

Our key hypothesis is that drawing attention to overdraft costs reduces demand for overdrafts. This would hold if consumers tend to underestimate overdraft costs and have limited/reactive attention, in which case mentioning price could jog memory that overdrafts are costly, motivate individuals to learn the correct and higher price, and/or increase attention to avoiding overdrafts. In contrast, mentioning price will not reduce overdraft demand if consumers are classically rational and hence correctly perceive overdraft costs on average.

Our test of this hypothesis is one-sided in the sense that in our experiment YK mentions price only while also cutting it: the overdraft pricing arms of our experiment all offer 50% discounts on overdrafts. This stacks the deck against finding a negative effect of mentioning overdraft price, since even a behavioral consumer who is susceptible to shrouding presumably prefers lower prices. Yet, we still find that mentioning overdraft price lowers overdraft demand. For example, the likelihood of overdrafting during the experiment is 1.2 percentage points lower ($SE = 0.4$ percentage points) for those receiving the discount offer relative to those receiving a message that mentions overdraft without also mentioning its price, on a baseline likelihood of 31%.

Another behavioral hypothesis is that bundling the overdraft discount with a discount on debit card or automated bill payment usage further depresses demand. The idea is that mentioning overdrafts together with transactions that could trigger overdrafts could be a particularly powerful reminder to avoid such transactions, akin to the attention-by-association findings in Stango and Zinman (2014).³ Again, both classical and mechanical forces work against finding support for this hypothesis; in the absence of behavioral factors, the bundled discounts should generate weakly more outflows from the checking account, producing weakly more variance in the checking account balance and hence weakly more overdrafts. Nevertheless, we continue to find support for the behavioral hypothesis. Offering the overdraft discount alone reduces overdraft likelihood by only 0.7 percentage points ($SE = 0.5$) relative to messages that mention overdraft without also mentioning price, while the bundled reductions are 1.4 percentage points for automated bill payment ($SE = 0.5$) and 1.7 percentage points for debit card ($SE = 0.5$).

Importantly, discounts on the bundled products themselves do not backfire in the same way that overdraft discounts do: offering a discount for debit card usage or automated bill payment authorization weakly increases demand for those services. These results further highlight the distinction between advertising shrouded attributes (overdraft) versus unshrouded attributes (debit or autopay).

As noted above, our experimental design also includes an overdraft availability promotion that does not mention price. We can identify its effects on demand

³ Stango and Zinman (2014) find that drawing consumer attention to spending control, monitoring account balances, or other bank fees leads to overdraft reductions.

by comparing it to messages that do not mention overdraft at all (they promote debit or automated bill payment transactions alone). We find that the overdraft availability message *increases* overdraft likelihood, by about 0.9 percentage points ($SE = 0.4$). We do not find heterogeneous effects based on prior overdraft experience (although the confidence intervals do not rule out meaningful differences), suggesting that overdraft availability is not salient/top-of-mind even if the consumer is already informed about availability in some classical sense. This finding suggests that, after acquiring a customer, firms can promote nonprice aspects of add-ons without drawing attention to costs.

Together, our results are consistent with models of shrouded equilibrium and limited/reactive consumer attention. In particular, they support (1) the key modeling assumption that consumers tend to underestimate add-on costs (if consumers' estimates were unbiased, then offering a discount would weakly increase demand), (2) the key assumption that firms lack incentives to unshroud prices, and (3) a key prediction of reactive attention models that consumers respond differently when advertising highlights different add-on attributes (price or availability). Shifts from shrouded to unshrouded equilibria may therefore be costly to generate and sustain. In Section V we discuss alternative, more classical interpretations, and why they do not provide as complete an explanation of our setting and results.

Our paper informs several other literatures on limited attention, salience, and advertising. First, our results provide insights into what comes to mind and what does not (Eliaz and Spiegler (2011), Bordalo, Gennaioli, and Shleifer (2013), Gabaix (2014), Hanna, Mullainathan, and Schwartzstein (2014), Karlan et al. (2016)). Second, our results show that price promotions have attention effects that can backfire from the promoter's perspective, thereby adding evidence to literatures on the psychology of incentives (Gneezy and Rustichini (2000), Kamenica (2012)) and price changes (Hastings and Shapiro (2013)) that have not yet focused much on shrouded prices. Our results are consistent with findings from other domains suggesting that consumers respond differently to base prices and add-on prices (Chetty, Looney, and Kroft (2009), Brown, Hossain, and Morgan (2010), Anagol and Kim (2012)).⁴ Lastly, our results are consistent with predictions from "transaction utility" models in which perceptions of fairness, not merely absolute value of a product or service, influence willingness-to-pay (e.g., see Thaler (1983)).

Relatedly, our findings contrast with those in Ferman (2016) and Elizondo and Seira (2017), who find little impact of messaging in Brazil and Mexico, respectively, that makes the *base* price of high-interest credit cards more prominent. Our results are broadly consistent with prior work that shows that advertising content can have important and surprising effects on decisions about expensive debt (Bertrand et al. (2010)), and that messaging from banks

⁴ We do not actually observe price sensitivity to the base price in our setting. But, given our result that cutting overdraft prices depresses overdraft demand, we can infer differential sensitivity to base and add-on prices simply by assuming that cutting the base price would not decrease demand for checking accounts.

can change the behavior of existing customers (Cadena and Schoar (2011), Karlan, Morten, and Zinman (2015), Kast, Meier, and Pomeranz (2016)). Our results on the long-run effects of short-run messaging complement the literature on the dynamics of learning and/or attention regarding add-on charges (Haselhuhn et al. (2012), Agarwal et al. (2013), Ater and Landsman (2013), Stango and Zinman (2014)), may help explain why advertising is so prevalent (treatment effects dissipate quickly and hence repeated exposure matters), and suggest that short-run behavior changes do not necessarily induce learning or greater sophistication about attention (Schwartzstein (2014), Manoli and Turner (2015)).⁵

The rest of the paper is arranged as follows. Section I describes the overdraft market in our study site, Section II explains our experimental design and describes our sample. We present our empirical results in Section III, and provide a detailed discussion in Section IV. Finally, Section V concludes. The Appendix provides figures and tables with additional results.

I. Setting: The Overdraft Market in Turkey

We discuss the economic importance of overdrafts in the introduction. Here, we describe our partner bank, YK, and the Turkish overdraft market. Our setting has many similarities to overdraft markets in other countries, like the United States and United Kingdom, with a key difference being substantially lower prices in Turkey (60% APR versus \$25 to \$40 per-transaction fees for very small and short-term loans).⁶

A. Retail and Mobile Banking in Turkey

Turkey's retail/consumer banking industry is concentrated. Only about 30 banks are licensed to take deposits, and the five largest banks have greater

⁵ Stango and Zinman, (SZ, 2014) is probably the empirical paper most closely related to ours. SZ use quasi-experimental variation in survey content in a market research panel in the United States to identify effects and dynamics of attention to overdraft fees. Similar to here, SZ find that an attention shock mentioning overdraft costs reduces overdraft usage, and that repeated attention shocks cumulate to some extent, although they depreciate more quickly in our setting. Aside from the obvious differences between the two study designs—market research surveys versus bank advertising as attention shocks, quasi-random versus random variation, United States versus Turkey—there are at least two other key differences. First, we have randomly assigned price variation. Second, our treatments include some messages that mention the overdraft service but not its cost. These differences lead to the surprising new inferences that bringing overdrafts to mind increases demand, but bringing the price of overdrafts to mind, even if accompanied by a discount, lowers demand. See also Liu, Montgomery, and Srinivasan (2014) and Hunt, Kelly, and Garavito (2015).

⁶ For details on overdraft markets in the United States and Europe, see, for example, Stango and Zinman (2014) and various government reports (General Accounting Office (2008), Bakker et al. (2014), Competition and Markets Authority (2014), Financial Conduct Authority (2014)). For policy developments in the United States and European Union, see, for example, FDIC Supervisory Guidance warning banks about the risks of “excessive use” of overdrafts by customers and “maximizing fees” by banks, Dougherty (2014), and the European Parliament's Directive on Payment Accounts (issued April 2014).

than 50% market share. YK is in the top five based on both total assets and number of branches, and is publicly traded. In recent years, the Turkish market has become known for innovation in retail banking, with one bank rolling out the largest biometric ATM network in the region, and another bank becoming the first in the world to make money transfers possible on Facebook. Turkey has the highest rate of mobile banking in Europe among Internet users, at around 50%, according to a 2013 ING survey, and 91% of Turkish adults have a cell phone, for one of the highest penetration rates in the world.

B. Overdraft Practices

The focus of our experiment is a checking account overdraft product, which YK brands the “Flexible Account,” with features that were standard in Turkey (and throughout the world) during our study period. The product is an unsecured line of credit that allows qualifying customers to overdraw their account (i.e., to hold a negative balance) at a cost of 60% APR on outstanding credit (about 50% after adjusting for inflation). Credit card APRs tend to be substantially lower. The bank approves about 55% of checking account customers for the overdraft feature, which credit limits that vary based on underwriting but that are typically lower than other unsecured credit products. Although individuals can apply for the overdraft feature, which typically (and in our study) the bank automatically chooses individuals to be given the feature, which thus requires individuals to opt-out if they do not want it rather than opt-in if they do. Customers then use the line, automatically, any time their checking account falls (further) below zero. Negative balances begin accruing interest immediately. By law, any inflow to the checking account is automatically allocated first toward paying off overdraft credit. If inflows are not sufficient to clear the balance by the end of the statement date (four weeks), the bank sends the customer a notice and gives her about two weeks to pay at least the accrued interest. If the customer fails to make the required payment after 60 days, the bank freezes the overdraft line.

Overdrafts have attracted regulatory scrutiny because they—and their prices—are rarely featured at the customer acquisition stage, and often are not fully disclosed even postacquisition. Preacquisition, we are not aware of any mass marketing campaigns promoting overdraft usage during our sample period in Turkey. Regulators have found that, when banks did communicate with (prospective) consumers, they often failed to provide mandated disclosures or did so only perfunctorily.

Communications with existing customers also indicate a (partially) shrouded equilibrium during our sample period. Customers lacked easy access to information on their own overdraft usage and charges. Banks did not routinely issue monthly statements, due to the unreliable postal system, and electronic notifications of overdraft usage did not mention the interest rate. Electronic banking was and is the dominant channel for customers getting information on their accounts, and online information on overdraft usage was folded into the customer’s main transaction record rather than disaggregated. The

transaction record would specify the amount charged in overdraft interest, but not the interest rate. Landing pages—the first screen the customer sees after logging in online or at an ATM—would obscure whether overdraft was being used by highlighting the customer’s balance based on “available-to-withdraw” (including credit line) instead of just the checking account balance. A customer seeking interest rate information would need to either navigate through several different screens or contact a customer service representative. Banks promoted overdraft usage to existing customers in direct messaging campaigns, but those campaigns did not mention the level of the interest rate even when offering discounts on that rate; for example, an ad would say “half off your interest charges” instead of “half off your 60% interest rate.”

C. Overdraft Users and Usage in Turkey

Who overdrafts? In Turkey there are not much data on the characteristics of overdrafters (e.g., our data lack information on education or income), but there are some clues. Over half of Turkey’s population is unbanked, according to a 2012 World Bank report. Moreover, many checking account holders are not approved for overdraft lines of credit due to credit risk that banks cannot price. These facts together suggest that overdrafters come from the upper half of the income distribution in Turkey, although not from the uppermost percentiles, who presumably have wealth and access to cheaper credit that would tend to render overdrafting unlikely and/or relatively unattractive.

Our customer sample overdrafts frequently in the 10 months after our experiment started, despite having been selected for the experiment based on infrequent overdraft activity prior to the experiment (Section III. B; Table I). In any given month, 15% to 24% of our sample overdrafts and 45.8% of our sample overdrafts at least once between September 1, 2012 to June 30, 2013, paying a mean finance charge of 30.82TL (1TL = \$0.56 USD during our sample period) over the 10 months, (the 95th percentile is 228.08TL).

The sharp increase in overdrafting from baseline to (post-) experiment is probably not due to mean reversion, as overdrafting tends to be strongly serially correlated month-to-month. Rather, YK and its regulators attribute the sharp increase to expansionary monetary policy that encouraged consumer lenders to promote durable financing around the end of Ramadan and that in turn produced overdrafts as many borrowers struggled to manage their new debt service. In any case, the sharp increase in overdrafting is evident in both aggregate data and YK’s customer base. YK held back a no-message group of 39,000 customers from our experiment, and Figures 2(A) and (B) show similar trends in overdraft usage for this group compared to our experimental group of 108,000 customers.⁷

⁷ YK applied somewhat different and not entirely reproducible filters in selecting the no-message comparison group, so we do not use this group as a pure control group for analysis purposes. Creating a pure, randomized control group was not a point of emphasis at the design and implementation stages of the experiment, since the equilibrium was one of extensive direct messaging and hence the mapping of message versus no-message comparisons to theory is less clear than comparisons across different promotions.

Table I
Baseline Variables for Sample Description and Orthogonality Checks

Columns (1) to (10) report proportions, with standard errors in parentheses. Column (11) reports the p -value from the test of the joint hypothesis that the coefficients on the treatment indicators are zero. Variables are measured in July 2012 unless noted otherwise. Figure 1 provides details on the treatment arms. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

September 15 and Subsequent Messages										
Full Sample	Got Aug 30 Msg	Overdraft			Interest Discount		Auto Bill-Pay		Debit Card Discount	
		Any	Subsequent Msgs	w/Auto Bill-Pay Discount	w/Debit Discount	Subsequent Msgs	Subsequent Msgs	Subsequent Msgs	Subsequent Msgs	p -Value from Regress Row on Columns
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Female	0.289 (0.001)	0.288 (0.002)	0.288 (0.002)	0.289 (0.003)	0.288 (0.003)	0.288 (0.003)	0.288 (0.004)	0.288 (0.003)	0.288 (0.004)	1.000
Lives in Istanbul	0.275 (0.001)	0.275 (0.002)	0.275 (0.002)	0.275 (0.003)	0.274 (0.003)	0.275 (0.003)	0.274 (0.004)	0.276 (0.003)	0.276 (0.004)	1.000
Lives in Ankara	0.090 (0.001)	0.090 (0.001)	0.091 (0.002)	0.090 (0.002)	0.090 (0.002)	0.091 (0.002)	0.090 (0.003)	0.089 (0.002)	0.089 (0.003)	1.000
Lives in Izmir	0.050 (0.001)	0.050 (0.001)	0.050 (0.001)	0.050 (0.002)	0.050 (0.002)	0.050 (0.002)	0.050 (0.002)	0.050 (0.002)	0.051 (0.002)	1.000
Lives in Bursa	0.057 (0.001)	0.057 (0.001)	0.057 (0.001)	0.057 (0.002)	0.058 (0.002)	0.057 (0.002)	0.057 (0.002)	0.057 (0.002)	0.057 (0.002)	0.998
Married	0.572 (0.002)	0.572 (0.002)	0.572 (0.003)	0.572 (0.004)	0.571 (0.004)	0.571 (0.004)	0.571 (0.005)	0.573 (0.004)	0.573 (0.005)	1.000
Overdraft Limit between 1/2 and 1 monthly min wage	0.284 (0.001)	0.284 (0.002)	0.284 (0.002)	0.282 (0.003)	0.285 (0.003)	0.284 (0.003)	0.283 (0.004)	0.284 (0.003)	0.284 (0.004)	1.000
Overdraft Limit > monthly min wage	0.081 (0.001)	0.081 (0.001)	0.081 (0.001)	0.081 (0.002)	0.080 (0.002)	0.082 (0.002)	0.082 (0.003)	0.082 (0.002)	0.082 (0.002)	0.999
Main Act Bal: Balance below med	0.428 (0.002)	0.428 (0.002)	0.428 (0.003)	0.428 (0.004)	0.427 (0.004)	0.427 (0.004)	0.427 (0.005)	0.428 (0.004)	0.428 (0.005)	1.000

(Continued)

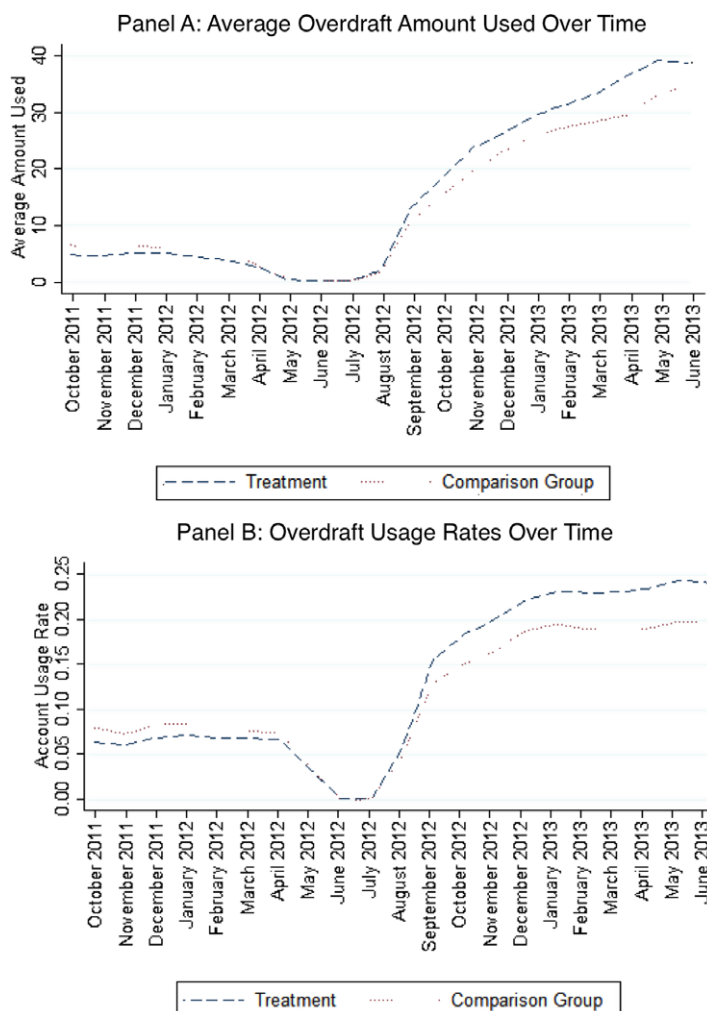


Figure 2. Overdraft usage over time. This figure plots average overdraft amounts used over time (A) and overdraft usage rates over time (B). “Overdraft Usage” is indicated by any overdraft balance in a given month. The “Treatment” group comprises the 108,000 accounts included in our experiment; the “Comparison” group was selected by YK and not included in our experiment. The experimental period covers August 30, 2012 through December 31, 2012. Dips just prior to the experimental period are due to the sample selection criterion of “no overdraft use in May, June, and July 2012.” (Color figure can be viewed at wileyonlinelibrary.com)

D. Policy Postscript

As noted above, overdraft practices have been attracting legal scrutiny around the globe. In July 2013, seven months after our experiment ended, the Turkish Competition Authority found that banks were benefiting from substantial markups over risk-based prices and fined 12 banks for price-fixing on

loans, including overdrafts. On May 27, 2013, the Turkish Central Bank imposed a binding price ceiling on overdraft APR. Turkish regulators have also focused on marketing and communications since our experiment ended, pressuring and working with banks to make overdraft usage, pricing, and costs more transparent.

E. Ethics of Experimenting with High-Cost Credit

In seminars we are frequently asked whether researchers should partner with a lender that is seeking to sell more high-interest rate loans. We think yes, in this circumstance as in many related experiments on microcredit where take-up is an outcome of interest, for four main reasons. First, an ethical concern here presumes that high-cost consumer credit harms consumers. But extensive research on this question suggests that a different assumption is warranted—(weakly) beneficial impacts for consumers (Karlan and Zinman (2010), Zinman (2014), Banerjee, Karlan, and Zinman (2015)). Second, YK's advertising was truthful and its terms were competitive. Thus, combining the first and second points, the experiment was not trying to convince consumers to accept a bad deal either in absolute terms or compared to market alternatives. Third, YK was going to promote overdraft usage among its existing customers with or without the participation of the research team and we helped convince bank management to feature prices despite its skepticism about the effectiveness of past overdraft price promotions. Fourth, YK and the research team contracted ex ante that the academic coauthors would have unrestricted intellectual freedom to report the results and disseminate them publicly to benefit regulators and further scientific knowledge.

II. Experimental Design, Sample, and Data

A. Experimental Design and Implementation

Figure 1 summarizes the experimental design and details the script of each message variant. The field experiment randomly assigns message content, frequency, and duration, as well as promotional offers on overdraft, debit card, and automated bill payment, to a sample of 108,000 checking account holders. YK did not send this sample any other promotional communications during this campaign. The only other communications YK sent to this sample were monthly account statements. YK sent the messages by SMS, which is the most common way banks communicate with their clients in Turkey.

YK began the experiment by sending half of the sample an “Overdraft Availability” message on August 30, 2012 that mentions the overdraft service and credit line but nothing about its cost.⁸ This first randomization is not crucial for testing our main hypotheses; it served primarily as a pilot for the subsequent

⁸ “We remind you that, for your immediate cash needs, you have a Flexible Account at Yapi Kredi with [custom fill] TL limit. Have a nice day.” One might think of this message as a “reminder” because the bank's policy and Turkish law require upfront disclosure of the overdraft features and

randomizations and a test for a heterogeneous treatment effect suggested by some of the motivating theoretical models (Section IV).

YK continued the experiment on September 15, 2012 with a second, independent randomization, sending each person in the sample one of six randomly assigned promotions. Half of the sample received one of the three “Overdraft Interest Discount” messages detailed in Figure 1, with one-third of this group (or of this half) getting only an overdraft discount, one-third getting an overdraft discount plus an automated bill payment discount, and one-third getting an overdraft discount plus a debit card discount. The other half of the sample did not get an overdraft interest discount, with one-third of this half getting the automated bill payment discount only, one-third getting the debit card discount only, and one-third getting only the overdraft availability reminder message described above.

An independent frequency randomization determined whether YK resent the September 15 message frequently (every 10 days), less frequently (every 20 days), or not again during the campaign period. A campaign duration randomization then determined whether the price promotion(s) or overdraft availability reminder, and any repeated messaging subsequent to September 15, lasted until November 15 or December 15.

Note that YK sent at least one message to everyone in our sample during the experiment. YK preferred this design feature because, like other banks, it often sends promotional and reminder messages to its customers and thought it would be counterproductive to scale back directed advertising to zero. The research team also preferred this design—placebo communications with respect to overdraft, rather than pure control—because any contact from YK could trigger the customer’s attention and affect her usage of YK products.⁹

B. Baseline Data on Sample Characteristics, and Balance Checks

YK sought to promote overdraft usage among existing clients who it deemed most likely to be close to the margin of overdrafting. To this end it selected customers for the experiment based on the following criteria: owned a YK checking account for at least a year, were in good standing in the account, and had a debit card linked to it; had an active cell phone; maintained an average total deposit account balance <5,000TL over the three months prior to the start of the experiment (May to July 2012); did not have more than three automated bill payments set up; and already had the overdraft service in place but had not used it during the prior three months. Many of these customers did have some experience with the product before the three-month period prior to the experiment. Our pretreatment data go back as far as September 2011, and from

pricing. However, given that the service is offered on an opt-out basis, and that our motivating questions concern shrouding prices, we allow for the possibility that this message provides new information rather than being a simple reminder. We explore this possibility in Section IV.

⁹ A closely related way of framing our interest in placebo communications is that frequent direct-messaging is the equilibrium, but we thought that the experiment would link more tightly to theory if the only deviations from equilibrium were around (partial) unshrouding.

September 2011 to April 2012 18.4% of our sample overdrafted at least once, with an average daily overdraft balance of 4.42TL (*SD* 23, *Max* 940) among these account holders.

Table I summarizes the baseline data available to us (column (1)) and checks balance across treatment assignments (columns (2) to (10)). In terms of demographics, we have information only on gender (29% female), city of residence (28% Istanbul, 23% outside the four largest cities), and marital status (57% married). This information is collected by the bank at the account-opening stage and can be updated later by the client. Besides pretreatment data on overdraft usage (described above and in Table I), we also have data on the other behaviors targeted by the experiment: debit card usage and automatic debits for bill payments. We stratified on each of these baseline variables. The last column confirms that none of them is correlated with treatment assignment by regressing each row variable on the treatment assignments indicated in columns (2) to (10).

C. Follow-Up Data

YK provided us with data on overdraft usage, debit card usage, automated bill payment authorizations, and deposit account balances, at the account-month level, from September 1, 2012 through the end of June 2013 (although we do not use the June 2013 data due to the binding price cap imposed at the end of May 2013). In addition to the monthly data, YK provided us with daily data on overdraft usage for the experimental period, September 15 to December 15, 2012. We use these data to construct outcome variables for estimating the short-run and longer-run treatment effects detailed below.

D. Hypotheses and Tests

Our main hypothesis is that drawing attention to the cost of overdrafting will depress demand for it. The motivation for this hypothesis comes from two key features of behavioral models of costly add-ons like overdrafts. First, consumers tend to underestimate, and have limited/reactive attention to, add-on costs.¹⁰ Mentioning price could thus jog memory that overdrafts are costly, motivate individuals to learn the correct and higher price, and/or increase attention to

¹⁰ Consumers can underestimate the price conditional on overdrafting (perhaps because they anchor on substantially lower prices for other bank services or sources of short-term credit), and/or underestimate the likelihood of overdrafting (perhaps they perceive it to be zero because they assume the bank will not let them overdraft, or perhaps they are overconfident about avoiding overdrafts). Grubb (2015) assumes that consumers perceive the price accurately but underestimate the likelihood of overdrafting, because they underestimate their cost of attention to balances. See also Grubb (2009). In terms of the other models most closely related to our setup, Gabaix and Laibson (2006), Heidhues, Köszegi, and Murooka (2017), and Bordalo, Gennaioli, and Shleifer (2015) assume that some consumers underestimate a reduced-form add-on cost that, in our setting, is the product of the price of overdrafting and the expected number of overdrafts (or, on the extensive margin, the product of the price of overdrafting and the likelihood of overdrafting).

avoiding overdrafts. Any of these channels could lead the consumer to revise her cost estimate upward and reduce her overdraft usage. In contrast, drawing attention to overdraft cost will not reduce demand if consumers are classically rational and hence correctly perceive overdraft costs on average. Second, and closely related to the demand response in the first feature, unshrouding costs (e.g., by drawing consumer attention to them) may be unprofitable for the add-on supplier.

We test this hypothesis by comparing overdraft usage during the experimental period between when customers sent an Overdraft Interest Discount message and when customers sent an Overdraft Availability message. Both types of message promote overdraft usage, but only the Discount message says anything about the price. As noted in the introduction, this test actually stacks the deck toward rejecting our main hypothesis because the Discount message does not simply draw attention to overdraft costs, but also *cuts* the price in half.¹¹ As such, accepting the hypothesis that the Overdraft Interest Discount message reduces demand—despite offering a much lower price—would be an especially strong indication that add-on price advertising unshrouds add-on costs for consumers, leading consumers to sharply revise their cost perceptions upward, depressing demand for the add-on and proving unprofitable for the supplier.

A second behavioral hypothesis is that bundling the overdraft discount with a discount on debit card or automated bill-payment usage will depress demand further. The idea is that mentioning overdrafts together with transactions that can trigger overdrafts could be a particularly powerful reminder to avoid overdrafts, akin to the findings in Stango and Zinman (2014) that drawing consumer attention to spending control mechanisms to monitor account balances, or other bank fees leads to overdraft reductions. Again, both classical and mechanical forces work against finding support for the behavioral hypothesis: in the absence of inattention, the bundled discounts should generate weakly more outflows from the checking account, producing weakly more variance in the checking account balance and hence weakly more overdrafts.

We test this bundling hypothesis by comparing overdraft usage, during the experimental period between when customers sent an Overdraft Interest Discount only message and when customers sent an Overdraft Interest Discount message that also includes a discount for automated bill payment or debit card use.¹²

Third, we hypothesize that promoting overdraft availability without mentioning price will change demand. Classically, rational consumers will have

¹¹ It would have been theoretically and statistically desirable to have experimental arms that simply mentioned the price of overdrafting without cutting it, and/or that mentioned the regular price (60% APR) while cutting it, but YK deemed such messaging too far off-equilibrium to be viable.

¹² This is a conservative test of the hypothesized behavioral mechanism: the bundled messages are longer, and if longer messages tax limited attention, we would expect them to push treatment effects on overdrafting toward zero instead of further depressing demand. Ignoring a message makes it akin to getting no message at all.

accurate perceptions of overdraft availability, at least on average, and should not respond. In contrast, behavioral mechanisms could generate either a negative response if advertising availability brings overdraft costs to mind or a positive response. A positive response could arise if the overdraft service generally, and not just its cost, is not at top of mind for consumers. If consumers have reactive attention, then drawing attention to a positive feature of the service—its availability when one is short on cash—can increase demand for it. Another potential mechanism is that the message operates on consumers who thought that overdrafting was impossible and thus perceive zero likelihood and an infinite price of overdrafting. In this case, the availability message could increase demand if it reduces the perceived price by more than it increases the perceived likelihood. We note that this mechanism would still be consistent with consumers underestimating overdraft *costs* at baseline, since impossibility implies zero expected cost.

We test whether and how promoting overdraft availability affects demand by comparing overdraft usage among customers sent the Overdraft Availability message to customers sent messages that promote only debit card or automated bill-payment usage and do not mention overdraft at all.

Our final hypothesis focuses on understanding the dynamics of attention and overdraft behavior. The dynamics shed light on whether consumer learning and/or persistent attention to add-on costs break a shrouded equilibrium. For example, in Heidhues, Köszegi, and Murooka (2017) the profitability of high add-on prices is “limited by consumers’ ex post demand response to add-on prices” (p. 341), raising the possibility that, at the customer level, the firm wants to shroud at the customer acquisition stage but then unshroud while cutting the price of the add-on. In contrast, consumer forgetting and/or reactive attention to add-on costs increases the cost of debiasing consumers, flattens or even reverses standard demand responses, and can make a shrouded equilibrium more durable.

To test the dynamics, we examine data from the postcampaign period (January to May 2013). Treatment effects will persist if consumer learning or attention with respect to add-ons is durable. Treatment effects will not persist if consumers quickly forget about add-ons or only attend to them when induced to do so by external stimuli like advertising.

III. Specifications and Results

We estimate ordinary least squares (OLS) regressions at the level of YK’s randomizations—the checking account, indexed by i — and a time frame that corresponds to either during- or postexperimental advertising (indexed by t):

$$Y_{it} = T_i\beta + X_i\alpha + e_{it},$$

where Y is some measure of a behavior targeted by the marketing campaign (overdraft usage, debit card usage, or an automatic debit for bill payment). In Tables II and III, we measure outcomes over a time period designed to capture

Table II
Effects of Overdraft Marketing on Overdraft Usage during Experiment (September to December 2012)

Each column presents treatment effect estimates from a single OLS regression, with Huber–White standard errors (in parentheses), of the usage measure in the column heading on the experimental variables in the rows. Variables in rows (4) to (7) identify the additive effect of one or more of the interest discount arms, that is, they identify the difference in treatment effects between the Overdraft Availability message and the Overdraft Interest Discount message. Usage measures cover September 15 to December 31, 2012. Account balances are in Turkish lira, where 1TL = \$0.56 at the start of the experiment. Figure 1 summarizes the experimental design and shows message scripts. Regressions also include controls for randomization strata. The unit of observation is the unit of randomization: a checking account. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

LHS Variable:	Overdraft Account Used		Days with Overdraft Balance		Mean (Overdraft Account Balance)	
	(1)	(2)	(3)	(4)	(5)	(6)
August 30 th Message:						
(1) Overdraft Availability	0.0007 (0.0028) Omitted	0.0007 (0.0028) Omitted	0.0869** (0.0365) Omitted	0.0869** (0.0365) Omitted	1.0554** (0.4701) Omitted	1.0554** (0.4701) Omitted
(2) No Message						
September 15 Message:						
(3) Overdraft Mention: identifies effect of Overdraft Availability message	0.0089** (0.0042) -0.0124*** (0.0040)	0.0089** (0.0042)	0.1321** (0.0553) -0.1607*** (0.0522)	0.1321** (0.0553)	0.9562 (0.7029) -0.7427 (0.6718)	0.9562 (0.7029) -0.7427 (0.6718)
(4) Overdraft Mention, with Interest Discount						
(5) Overdraft Mention, with Interest Discount only		-0.0069 (0.0049)		-0.0854 (0.0640)		-0.3382 (0.8108)
(6) Overdraft Mention, w/Interest + Auto Bill-Pay Discounts		-0.0137*** (0.0049)		-0.1884*** (0.0635)		-1.0909 (0.8195)
(7) Overdraft Mention, w/Interest + Debit Card Discounts		-0.0167*** (0.0049)		-0.2083*** (0.0632)		-0.7989 (0.8397)
(8) No Overdraft Mention = Auto Bill-Pay Discount Only or Debit Card Discount Only	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted
Mean(LHS)	0.3077 (0.46)	0.3077 (0.46)	2.7676 (6.05)	2.7676 (6.05)	26.8511 (77.97)	26.8511 (77.97)
SD(LHS)	108,000	108,000	108,000	108,000	108,000	108,000
Observations						

Table III
Effects of Auto Bill-Pay and Debit Card Marketing during Experiment (September to December 2012)

Each column presents treatment effect estimates from a single OLS regression, with Huber–White standard errors (in parentheses), of the usage measure in the column heading on the experimental variables in the rows. Variables in rows (2) and (4) identify the additive effect of the overdraft interest discount, that is, they identify the difference in treatment effects between the discount on some other service in row (1) or (3) and that discount bundled with the overdraft interest discount. Usage measures cover September 15 to December 31, 2012. “POS” = point of sale. Account balances are in Turkish lira, 1TL = \$0.56 at the start of the experiment. Figure 1 summarizes the experimental design and shows message scripts. Regressions also include controls for randomization strata. The unit of observation is the unit of randomization: a checking account. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

LHS Variable:	Any Debit Card POS Transactions (1)	Any Auto Bill-Pay Registered (2)	Overdraft Account Used (3)	Days with Overdraft Balance (4)	Avg Overdraft Account Balance (TL) (5)
September 15 Message:					
(1) Debit Card Discount (includes both Debit Card Discount Only, and Overdraft Interest Discount + Debit Discount)	0.0053 (0.0040)		0.0029 (0.0049)	−0.0049 (0.0635)	−0.3401 (0.7896)
(2) Debit Card Discount + Overdraft Interest Discount	−0.0008 (0.0049)		−0.0127*** (0.0048)	−0.1180* (0.0627)	−0.1206 (0.8204)
(3) Auto Bill-Pay Discount (includes both Auto Bill-Pay Discount Only, and Overdraft Interest Discount + Auto Bill-Pay Discount)		0.0042*** (0.0011)	−0.0070 (0.0048)	−0.0883 (0.0634)	−0.8953 (0.7969)
(4) Auto Bill-Pay Discount + Overdraft Interest Discount		−0.0018 (0.0013)	0.0001 (0.0048)	−0.0147 (0.0629)	0.1426 (0.8060)
(5) Sep 15 Message: Overdraft Interest Discount only	Omitted	Omitted	Omitted	Omitted	Omitted
Controls for Aug 30 and Sep 15 Overdraft Availability Only Msgs	Yes	Yes	Yes	Yes	Yes
Mean(LHS)	0.3367 (0.47)	0.0134 (0.11)	0.3077 (0.46)	2.7676 (6.05)	26.8511 (77.97)
SD(LHS)	108,000	108,000	108,000	108,000	108,000
Observations					

Table IV
Effects of Overdraft Marketing on Overdraft Usage after Experiment
(January to May 2013)

Each column presents treatment effect estimates from a single OLS regression, with Huber–White standard errors (in parentheses), of the usage measure in the column heading on the experimental variables in the rows. Variables in rows (4) to (7) identify the additive effect of one or more of the interest discount arms, that is, they identify the difference in treatment effects between the Overdraft Availability message and the Overdraft Interest Discount message. Usage measures cover January 1 to May 31, 2012. Account balances are in Turkish lira, where 1TL = \$0.56 at the start of the experiment. Figure 1 summarizes the experimental design and shows message scripts. Regressions also include controls for randomization strata. The unit of observation is the unit of randomization: a checking account. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

LHS Variable:	Overdraft Account Used		Mean (Overdraft Account Balance)	
	(1)	(2)	(3)	(4)
August 30 th Message:				
(1) Overdraft Availability	−0.0004 (0.0029)	−0.0004 (0.0029)	0.7385 (0.6262)	0.7383 (0.6262)
(2) No Message	Omitted	Omitted	Omitted	Omitted
September 15 Message:				
(3) Overdraft Mention: identifies effect of	0.0055 (0.0044)	0.0055 (0.0044)	0.0320 (0.9344)	0.0320 (0.9344)
Overdraft Availability message				
(4) Overdraft Mention, with Interest Discount	−0.0050 (0.0042)		0.1192 (0.8916)	
(5) Overdraft Mention, with Interest Discount only		−0.0024 (0.0051)		0.8686 (1.1030)
(6) Overdraft Mention, w/Interest + Auto Bill-Pay Discounts		−0.0067 (0.0051)		−0.7063 (1.0724)
(7) Overdraft Mention, w/Interest + Debit Card Discounts		−0.0059 (0.0051)		0.1954 (1.1058)
(8) No Overdraft Mention =	Omitted	Omitted	Omitted	Omitted
Auto Bill-Pay Discount Only or Debit Card Discount Only				
Mean(LHS)	0.3713	0.3713	34.0032	34.0032
SD(LHS)	(0.48)	(0.48)	(103.98)	(103.98)
Observations	108,000	108,000	108,000	108,000

immediate/short-run treatment effects: t covers September 15 to December 31, since the bank sent everyone at least one promotional message starting September 15 and sent the last promotional messages on December 15. Table IV measures outcomes over the postexperiment period, January 2013 to May 2013.

T is a vector of treatment assignments (see Figure 1), where β is the vector of estimated coefficients on those treatment variables and X is a vector of the stratification variables used to block the randomization (see Table I).

Our main tables define the treatment vector to test our main hypothesis with the simplest possible presentation. In particular, we make use of additive

specifications that enable us to keep the omitted treatment category consistent across columns within each table and to facilitate inference about comparisons of theoretical interest: overdraft promotions that mention price versus those that do not (rows (4) to (7) in Tables II and IV) and price promotions on other banking services that are bundled with overdraft discounts versus those that are not (rows (2) and (4) in Table III). The Appendix presents a more comprehensive set of results.

A. Main Effects of Overdraft Promotions on Overdrafting, during Experiment

Table II presents the estimated effects of the different overdraft promotions on three measures of overdraft usage during the experimental period (September 15 to December 31, 2012).

Rows (1) and (3) estimate effects of the Overdraft Availability message, which does not mention costs or offer a discount. This message *increases* demand for overdraft relative to not getting a message on August 30th (rows (1) and (2)) and relative to messages that do not mention overdraft (row (3)).¹³ The magnitudes are on the order of 1 percentage point in 1/0 usage, 0.1 days with an overdraft balance, and 1TL in average overdraft balance. These results support the hypothesis that promoting overdrafts without reference to cost increases demand, the most straightforward interpretation of which is that consumers have limited and reactive attention to all aspects of the overdraft service, not just to its cost.

Row (4) presents estimates of the effect of getting an overdraft interest promotion relative to getting the Overdraft Availability message, which does not mention price. Reading across columns one can see that this effect is negative—offering a lower price leads to *lower* demand—with estimates for two of the three demand measures having p -value < 0.01 . The extensive margin decreases by 1.2 percentage points ($SE = 0.4$), and days with a balance falls by 0.16 ($SE = 0.05$), for declines of 4% and 6% relative to the sample means. These results support the hypothesis that drawing attention to overdraft costs reduced demand (even while offering a 50% discount!), with the key implications being that consumers tend to underestimate overdraft costs and banks lack incentives to unshroud or compete on overdraft prices because doing so backfires.¹⁴

Rows (5), (6), and (7) decompose the overdraft interest discount effects into the overdraft discount only (row (5)) and the overdraft discount bundled with discounts for autopay (row (6)) or debit card use (row (7)). These results suggest that the overdraft discount by itself does not depress demand to a statistically

¹³ One might wonder whether the “Have a nice day” portion of the availability message might be driving the effects rather than a reminder about availability per se, but if this were the case one might expect to see customers responding to the bank’s friendliness by using other banking services more. We do not find any such evidence (see Appendix Table A2, row (3)).

¹⁴ In Appendix Table A1, we confirm that lower demand likely maps into lower profitability for the bank, as we do not find any evidence that account balances increase to offset lost overdraft revenue with increased implicit interest. Stango and Zinman (2014) also infer that consumers do not engineer overdraft reductions with balance infusion.

significant degree (although each of the point estimates is negative); rather, it is the bundled discounts that drive the backfiring effect (see also Table III, columns (3) to (5), rows (2) and (4), which compare bundled discounts to the overdraft discount only). These results support the hypothesis that bundling overdraft discounts with other discounts is particularly demand-depressing. As discussed above, our favored interpretation is that attention to overdraft costs is reactive and associative; we consider alternative interpretations in Section V.¹⁵

The treatment effects in Table II are likely economically as well as statistically significant, for several reasons. First, they suggest that drawing attention to overdraft costs induces upward-sloping demand, which is rarely seen and hence qualitatively important. Second, the messaging here does not mention the level of costs; instead, YK offers to give back “half of the interest charges.” A pure debiasing strategy would likely mention the price or cost level, along the lines of: “Beware of overdrafts at 60% APR!” Messaging that highlighted the 60% APR might depress demand even more, suggesting that we identify a lower bound on the effects of unshrouding overdraft costs. Third, messaging costs are low, and hence bank strategy is sensitive to small changes in demand. Fourth, our estimates identify the effects of *sending* messages rather than of consumers *attending* to them. Some recipients may have ignored the messages. This implies that our estimates represent lower bounds for the effects on consumers who actually paid attention to the messages.

B. Do All Promotional Discounts Backfire? No

In Table III, we check whether other promotional discounts backfire as well by estimating treatment effects of the debit card and automated bill payment discount offers on their targeted behaviors during the experiment. The key results here are presented in columns (1) and (2). In particular, row (1)-column (1) shows that offering the debit discount alone weakly increase debit card usage by 0.5 percentage points ($SE = 0.4$), and row (3)-column (2) estimates that the autopay discount alone increases autopay signup by 0.4 percentage points ($SE = 0.1$). These effects are each scaled relative to the Overdraft Interest Discount Only message (notice that the omitted category is different here from in Table II, since here we are primarily interested in whether the other discounts affect demand for their services).¹⁶ Row (2)-column (1) and row (4)-column (2) test whether bundling a discount with the Overdraft Interest Discount message performs differently with respect to demand for debit cards and autopay,

¹⁵ Another possible mechanism is directly testable: in Appendix Table A1, Panel B, we do not find any evidence that the automated bill payment and debit card promotions lead to higher account balances, casting doubt on the possibility that these promotions motivate individuals to maintain larger bank account balances and thereby produce less overdrafting.

¹⁶ Alternative comparisons, not shown in the table, produce similar inferences. If, for example, we instead estimate effects relative to not getting a debit card discount, the Debit Card Discount Only coefficient for 1/0 debit card usage is 0.0067 ($SE = 0.0039$), and the Debit Card Discount + Overdraft Interest Discount coefficient is 0.0059 ($SE = 0.0039$).

than offering a discount on debit or autopay alone. We do not find evidence of differential effects.

In sum, we find no evidence that offering discounts on other bank services backfires with respect to demand for those services and some evidence that they increase demand as intended.

C. Do Treatment Effects Persist? Postexperiment Effects of Overdraft Messaging

Table IV reestimates our main specifications from Table II over the postpromotional campaign period for the two overdraft usage measures for which we have data from January to May 2013. (Recall that the most-intensively treated account holders in our experiment received their last message on December 15, 2012.) We find no statistically significant evidence that treatment effects persist over the five-month postexperiment period: the overdraft discount effect is no longer demand-depressing, and the overdraft availability effect is no longer demand-increasing. Figure 3, Panels A and B, plot the two treatment effects and their confidence intervals month-by-month for our 1/0 measure of overdraft usage. As can be seen, both treatment effects trend toward zero over the postcampaign period and lose statistical significance about two months after the campaign (between February and March).

The results in Table IV suggest that the overdrafting changes induced by bank promotions are not “sticky”: the changes do not persist long after the promotions stop. This could be related to the fact that consumers overdraft passively in the course of checking account usage (rather than explicitly drawing from a line of credit): overdrafts are plausibly low “exposure” and hence relatively likely to be ignored or forgotten (Bordalo, Gennaioli, and Shleifer (2015)). In any case, the lack of persistent effects suggests that consumer learning and/or attention concerning overdrafts depreciates quickly, and hence that advertising and debiasing campaigns must persist to be effective.

D. Heterogeneous Effects of Overdraft Messaging?

Our motivating models predict that responses to add-on advertising will vary with how well-informed and/or attentive the consumer is. For example, a well-informed and attentive consumer should exhibit a standard demand response to the overdraft interest discount (increasing, not decreasing, demand) and should respond weakly if at all to the overdraft availability reminder.

We construct two proxies for baseline exposure to the overdraft product and then test whether each proxy mediates our main treatment effects. The first proxy is recent overdraft use prior to the experiment. We find that 18% of our sample overdrafted at some point during September 2011 to August 2012. Table V interacts a prior use indicator with our main treatment variables and shows little evidence of heterogeneous treatment effects (p -values at the bottom of the table). The second proxy is generated by the August 30th Overdraft Availability message that YK sent to half of the sample. This initial

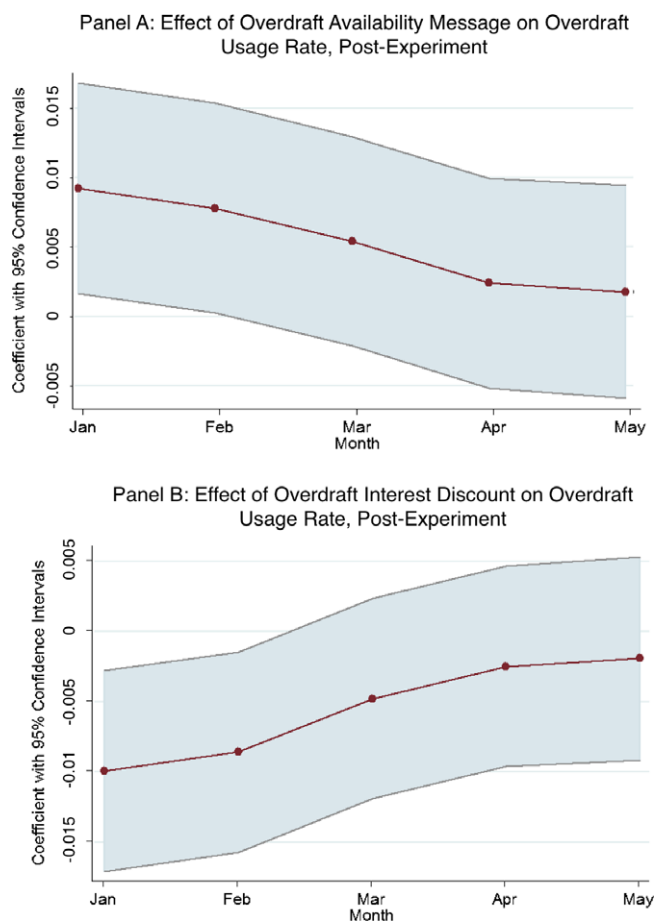


Figure 3. Effects of Overdraft Availability message and Overdraft Interest Discount on overdraft usage rate (A and B). This figure plots month-by-month treatment effects and associated confidence bands covering the postexperiment period (January to May 2013). Coefficients are estimated using a monthly version of the specification in Table IV, column (1). (Color figure can be viewed at wileyonlinelibrary.com)

message may have provided some consumers with information and/or drawn their attention to the add-on. But in Appendix Table A3 we show that while the point estimate for the interest discount is more negative for those who first received the August 30th Overdraft Availability message, the difference is not statistically significant (p -values at bottom of the table). We caution, however, that the confidence intervals for the heterogeneity results include both null and economically important point estimates.

We also examine heterogeneity with respect to baseline checking account balances. The intuition is that those with higher account balances may be less responsive to the messages (both availability and discount), as the messages are

Table V
Heterogeneous Treatment Effects by Prior Use

Each column presents treatment effect estimates from a single OLS regression, with Huber–White standard errors (in parentheses), of the usage measure in the column heading on the experimental variables in the rows. Variables in rows (4) to (11) identify the additive effect of one or more of the interest discount arms, that is, they identify the difference in treatment effects between the Overdraft Availability message and the Overdraft Interest Discount. Overdraft usage measures cover September 15 to December 31, 2012, except for prior use, which is measured 1/0 preexperiment over September 2011 to April 2012. Account balances are in Turkish Lira, where 1TL = \$0.56 at the start of the experiment. Figure 1 summarizes the experimental design and shows message scripts. Regressions also include controls for randomization strata. The unit of observation is the unit of randomization: a checking account. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

LHS Variable: Measured during Experiment: Over September 15 to December 31, 2012	Overdraft		Days with Overdraft Balance		Mean (Overdraft Account Balance)	
	(1)	(2)	(3)	(4)	(5)	(6)
(1) Prior Overdraft Use	0.3201*** (0.0067)	0.3201*** (0.0067)	4.0976*** (0.1093)	4.0976*** (0.1093)	41.6837*** (1.5148)	41.6835*** (1.5148)
No Overdraft Use September 2011 to April 2012	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted
September 15 th Message:						
(2) Overdraft Mention * Prior Overdraft Use	0.0193* (0.0107)	0.0193* (0.0107)	0.2546 (0.1807)	0.2546 (0.1807)	0.5915 (2.5326)	0.5920 (2.5326)
(3) Overdraft Mention * No Prior Overdraft Use	0.0055 (0.0044)	0.0055 (0.0044)	0.0912* (0.0521)	0.0912* (0.0521)	0.8841 (0.6385)	0.8843 (0.6385)
(4) Overdraft Mention, with Interest Discount * Prior Overdraft Use	−0.0205** (0.0101)		−0.2640 (0.1703)		0.4073 (2.3965)	
(5) Overdraft Mention, with Interest Discount * No Prior Overdraft Use	−0.0094*** (0.0041)		−0.1223*** (0.0492)		−0.8171 (0.6130)	
(6) Overdraft Interest Discount, No Other Discount *		−0.0176 (0.0124)		−0.1611 (0.2097)		1.1800 (2.9235)
(7) Overdraft Interest Discount, No Other Discount * No Prior Overdraft Use		−0.0036 (0.0051)		−0.0559 (0.0603)		−0.5237 (0.7335)
(8) Overdraft Interest Discount + Auto Bill-Pay Discount * Prior Overdraft Use		−0.0204 (0.0124)		−0.3532* (0.2093)		−1.6837 (2.9142)

(Continued)

Table V—Continued

LHS Variable: Measured during Experiment: Over September 15 to December 31, 2012	Overdraft Account Used		Days with Overdraft Balance		Mean (Overdraft Account Balance)	
	(1)	(2)	(3)	(4)	(5)	(6)
(9) Overdraft Interest Discount + Auto Bill-Pay Discount * No Prior Overdraft Use		−0.0109** (0.0050)		−0.1362** (0.0598)		−0.7935 (0.7550)
(10) Overdraft Interest Discount + Debit Card Discount * Prior Overdraft Use		−0.0236* (0.0124)		−0.2784 (0.2093)		1.7192 (3.0806)
(11) Overdraft Interest Discount + Debit Card Discount * No Prior Overdraft Use		−0.0137*** (0.0050)		−0.1745*** (0.0592)		−1.1335 (0.7528)
No Overdraft Mention = Auto Bill-Pay Discount Only or Debit Card Discount Only	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted
<i>p</i> -Value on <i>F</i> -test of equality between rows (2) and (3)	0.2348	0.2348	0.3852	0.3851	0.9108	0.9109
<i>p</i> -Value on <i>F</i> -test of equality between rows (4) and (5)	0.3068		0.4239		0.6203	
<i>p</i> -Value on <i>F</i> -test of equality between rows (6) and (7)		0.2952		0.6297		0.5716
<i>p</i> -Value on <i>F</i> -test of equality between rows (8) and (9)		0.4793		0.3190		0.7675
<i>p</i> -Value on <i>F</i> -test of equality between rows (10) and (11)		0.4578		0.6330		0.3678
Mean(LHS)	0.3077	0.3077	2.7676	2.7676	26.8511	26.8511
SD(LHS)	(0.46)	(0.46)	(6.05)	(6.05)	(77.97)	(77.97)
Observations	108,000	108,000	108,000	108,000	108,000	108,000

less relevant for them. As reported in Table VI, we do not find any evidence of heterogeneity in this regard. This could reflect the fact that most accounts, even those with above-median balances at baseline, dip low enough that overdrafting is viable and relevant. (Unfortunately, we lack baseline data on *minimum* balances.)

Another interesting margin of heterogeneity concerns the hypothesis that the most-intense overdraft users should be better informed about overdraft costs and more attuned to overdraft availability, in which case both treatment effects should attenuate and even flip sign at high usage. We examine this hypothesis by estimating quantile treatment effects, and do see hints of the expected pattern (Appendix Figure AI). However, we caution that these quantile regressions provide only coarse tests of the relationship between overdraft intensity and demand responses—they are underpowered, and they identify the heterogeneity of interest only with the additional assumption that ordinal position in the overdraft distribution is unaffected by the messages.

E. Other Treatment Variations: Messaging Frequency and Duration

Appendix Table A4 examines our messaging frequency treatments. We find that more frequent messaging amplifies the demand-depressing effect of the Overdraft Interest Discount message and the demand-increasing effect of the Overdraft Availability message.

The odd columns in Appendix Table A4 estimate the effect of one-shot versus repeated messaging for the September 15th message. In each treatment, the results are null for the one-shot and strong when the messages are repeated, that is, more negative for the discount and more positive for availability. Thus, the results in Table II are driven by repeated messaging, not the one-shot message. Four of the six comparisons between the one-shot and repeated messaging coefficients are statistically different from each other, and none of the six *p*-values are greater than 0.17. The even columns further separate repeated messaging into more versus less frequent (every 10 days versus every 20 days messaging) and, although the point estimates on more frequent messaging are larger in absolute value in five of the six comparisons, no difference is statistically significant. The key takeaways from Appendix Table A4 are that one message is not enough to generate an effect and that repeat messaging influences demand, with diminishing marginal effects from messaging every 10 versus 20 days.

Appendix Table A5 examines our other margin of messaging and promotional intensity, namely, the “duration” or length of time, over which the bank continued to send messages and offer discounts. Recall that short-duration campaigns lasted until November 15th, while long-duration campaigns continued until December 15th. We find little evidence that duration alone affects demand, either for the overdraft interest discount (row (5) versus row (6)) or for the overdraft availability message (row (2) versus row (3)), with the lone exception being the availability messaging effect on overdraft average balance (column (6)).

Table VI
Heterogeneous Treatment Effects by Baseline Account Balance

Each column presents treatment effect estimates from a single OLS regression, with Huber–White standard errors (in parentheses), of the usage measure in the column heading on the experimental variables in the rows. Variables in rows (2) to (11) identify the effect of one of the interest discount arms interacted with the baseline account balance. Usage measures cover September 15 to December 31, 2012. Account balances are in Turkish lira, where 1TL = \$0.56 at the start of the experiment. Figure 1 summarizes the experimental design and shows message scripts. Regressions also include controls for randomization strata. The unit of observation is the unit of randomization: a checking account. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

LHS Variable:	Overdraft		Days with		Avg Overdraft	
	(1)	(2)	(3)	(4)	(5)	(6)
Overdraft Account Used						
(1) Baseline Account Balance Above Median	0.0112 (0.0071)	0.0113 (0.0071)	−0.2678*** (0.0926)	−0.2674*** (0.0926)	−1.6892 (1.1565)	−1.6870 (1.1565)
(2) Sep 15 Message: Overdraft Mention * Baseline Acct Bal Above Median	0.0107* (0.0059)	0.0107* (0.0059)	0.1244* (0.0692)	0.1244* (0.0692)	0.8753 (0.9368)	0.8752 (0.9368)
(3) Sep 15 Message: Overdraft Mention * Baseline Acct Bal Below Median	0.0082 (0.0060)	0.0082 (0.0060)	0.1573* (0.0861)	0.1573* (0.0861)	0.9519 (1.0263)	0.9519 (1.0263)
(4) Sep 15 Message: Overdraft Mention, with Interest Discount *	−0.0138** (0.0056)	−0.0138** (0.0056)	−0.1219* (0.0655)	−0.1219* (0.0655)	−0.6248 (0.8936)	−0.6248 (0.8936)
(5) Sep 15 Message: Overdraft Mention, with Interest Discount *	−0.0117** (0.0057)	−0.0117** (0.0057)	−0.2070** (0.0812)	−0.2070** (0.0812)	−0.6918 (0.9825)	−0.6918 (0.9825)
(6) Sep 15 Message: Overdraft Interest Discount; No Other Discount		−0.0095 (0.0069)		−0.0695 (0.0803)		−0.3711 (1.0767)
* Baseline Acct Bal Above Median		−0.0053 (0.0070)		−0.1199 (0.0996)		−0.3425 (1.1917)
(7) Sep 15 Message: Overdraft Interest Discount; No Other Discount		−0.0129* (0.0069)		−0.1216 (0.0800)		−0.5927 (1.0773)
* Baseline Acct Bal Below Median		−0.0151** (0.0069)		−0.2608*** (0.0985)		−1.2366 (1.2193)
(8) Sep 15 Message: Overdraft Interest Discount; Auto Debit		−0.0189*** (0.0068)		−0.1742** (0.0791)		−0.9077 (1.1189)
* Baseline Acct Bal Above Median		−0.0145** (0.0070)		−0.2401** (0.0986)		−0.4921 (1.2285)
(9) Sep 15 Message: Overdraft Interest Discount; Auto Debit		0.7714 (0.7897)	0.7661 (0.4151)	0.7661 (0.4151)	0.9561 (0.9598)	0.9560 (0.9560)
* Baseline Acct Bal Below Median		0.6883 (0.8172)		0.6939 (0.2727)		0.9858 (0.6923)
(10) Sep 15 Message: Overdraft Interest Discount; Debit Card		0.6509 (0.3077)	2.7676 (6.05)	0.6024 (0.2766)		0.8026 (0.8026)
* Baseline Acct Bal Above Median		0.3077 (0.46)		2.7676 (6.05)	26.8511 (77.97)	26.8511 (77.97)
(11) Sep 15 Message: Overdraft Interest Discount; Debit Card		107.337 (107.337)	107.337 (107.337)	107.337 (107.337)	107.337 (107.337)	107.337 (107.337)
Discount * Baseline Acct Bal Above Median						
Discount * Baseline Acct Bal Below Median						
Discount * Baseline Acct Bal Above Median						
Discount * Baseline Acct Bal Below Median						
p-Value on F-test of equality between rows (2) and (3)						
p-Value on F-test of equality between rows (4) and (5)						
p-Value on F-test of equality between rows (6) and (7)						
p-Value on F-test of equality between rows (8) and (9)						
p-Value on F-test of equality between rows (10) and (11)						
Mean(LHS) Sept to Nov						
SD						
Observations						

Promotional intensity depends on duration *and* frequency, and the even numbered columns in Appendix Table A5 shed light on these interactions for the overdraft discount (rows (7) to (10)) and the overdraft availability message (rows (2) to (4)). Comparing, for example, the most intense price promotion (row (7): long duration + messages after September 15) to the least intense promotion (row (10): short duration, no messages after September 15), we find lower demand for the most intense promotion in all three cases, with p -values on the difference of 0.07, 0.08, and 0.002.¹⁷ We see a similar pattern of results for overdraft availability marketing, with the point estimate on the most intense promotion (row (2): long duration + messages after September 15) substantially larger than that on the least intense promotion (row (4): September 15 message and no subsequent ones), with p -values of 0.07, 0.25, and 0.02.

All told, we infer that more intense promotions amplify both the demand-increasing effect of the Overdraft Availability message and the demand-depressing effect of the Overdraft Interest Discount message. It could be the case that one message is sufficient to change behavior *if noticed*, but any one message is noticed with low probability. Alternatively, it could be the case that one message is sufficient to trigger awareness (of a shrouded attribute), but subsequent messages are required to trigger action. This also implies that demand responds more conventionally to less intense overdraft price promotion.

IV. Discussion

Taken together, our results are consistent with models in which consumers have limited and reactive attention to add-ons like overdrafts, and suppliers respond by shrouding add-on costs. Specifically, it seems that overdraft costs and availability are not at the top of consumers' mind, and even when brought closer to top of mind they do not stay there for long. As such, recent behavioral models of add-on pricing, marketing, and usage capture key aspects of reality, with consumers that tend to underestimate add-on costs and react strongly but temporarily when their attention is drawn to add-on, and with firms that lack incentives to unshroud or compete on add-on costs.

Classically rational models can explain at most a subset of our results. In principle, one could rationalize the finding that a large price discount decreases overdraft demand with a wealth effect that is big enough to counteract the standard price effect. In practice, such a wealth effect seems implausible. First, the wealth effect would likely operate among those who actually overdraft and have the most wealth to gain, yet we find effects on the extensive margin. Second, the demand reduction does not persist after the promotions stop, which implies that any wealth effect would have to be transient and begs the question of why, since the overdraft discount is conditional on usage and hence does not

¹⁷ Note that the point estimates on one-shot messaging about the overdraft discount suggest weakly standard/downward-sloping price effects on demand (rows (1) and (5) in Appendix Table A4, and rows (9) and (10) in Appendix Table A5).

alleviate liquidity constraints. Third, a wealth effect should operate through nonoverdraft discounts as well, yet we do not find any evidence that debit card or automated bill payment discounts alone reduce overdrafting. Fourth, a wealth effect does not explain why overdraft availability messaging also affects demand.

Another potential explanation for the overdraft price discount backfiring, particularly in light of Johnson, Meier and Toubia (2015), is that consumers view the offer as “too good to be true,” that is, they mistrust YK. But mistrust would not readily explain our other key results on availability and lack of persistence. First, it is silent on why availability increases demand and begs the question of why discount-driven mistrust would dissipate almost immediately after the campaign ends. Second, it is not clear why consumers would mistrust the overdraft interest discount but not other deep discounts that prevail in equilibrium, like “free” checking and teaser rates on credit cards. Third, a mistrust channel need not be distinct from the behavioral mechanisms described above; indeed, Johnson, Meier, and Toubia (2015) find that some “households expect there to be hidden fees and cumbersome processes that are not compensated by the attractiveness of the offer.” Fourth, it is not clear why our sample would respond by *decreasing* demand for overdrafts rather than simply ignoring the offer: do consumers assume that hidden costs exceed the value of the discount? Fifth, if YK’s clients did think that YK was trying to trick them, we might expect them to reduce their demand for other YK services. Yet, we find no such evidence (see Appendix Table A2), and the point estimates of the effect of the overdraft discount on the number of active YK accounts are actually positive rather than negative. Sixth, several institutional differences between our experiment and Johnson, Meier, and Toubia’s (2015) make mistrust more important in their context. In particular, the offer in Johnson, Meier, and Toubia (2015) *was* too good to be true in the sense of being government-subsidized, and that offer was made by a mortgage servicer at a time when the servicing industry was known to be mistreating and scamming customers (e.g., Consumer Financial Protection Bureau (2013)).

Rational inattention, and/or high search costs as in Ellison (2005), could explain our results, but under assumptions that strike us as antithetical to those sorts of models. For example, instead of remembering new information, consumers quickly forget it. Perhaps more critically, consumers systematically underestimate costs and credit lines instead of having accurate (if noisy) perceptions. But does it make sense to think of consumers as rational if they hold biased perceptions of contract terms in equilibrium? Much of behavioral economics answers this question with an emphatic no—indeed, it draws the line between rational versus behavioral based on a distinction between mean-zero versus biased deviations from classical assumptions about decision-making.

Relatedly, one could rationalize the bundled discount results as due to reduced attention costs more broadly, rather than to increased attention to add-on costs in particular, as the key factor mediating consumer choices. Specifically,

inducing use of autopay or debit cards could increase the customer's engagement with the bank, thereby lowering costs of monitoring her cash flows and reducing overdrafts.¹⁸ But messages that discount *only* the bundled products and do not mention overdraft do *not* reduce overdraft usage. Thus, the mechanism seems unlikely to hinge only on a (rational) increase in attention to the checking account more broadly. Some heightened awareness of add-on costs is likely key.

Notwithstanding the above, we are not dismissing rational or near-rational explanations for our results. Rather, we are merely speculating that behavioral models of limited attention, memory, and shrouding have great potential to explain the full picture.

V. Conclusion

Working with a large Turkish bank to test SMS direct marketing promotions to 108,000 existing checking account holders, we find that messages promoting a 50% discount on the overdraft interest rate *reduce* overdraft usage. In contrast, messages that mention overdraft availability without mentioning price *increase* usage. Neither change persists long, however, after messages stop. We also find some evidence that messaging intensity reinforces the main effects of overdraft discount and availability advertising—messaging more about the overdraft *discount* further reduces demand, while messaging more about overdraft *availability* further increases demand—and that messages offering discounts on debit card or autopay use along with overdraft backfire more than simply offering a discount on overdrafts. But not all discounts backfire: we find some evidence that debit card and autopay discounts increase usage of those features.

Our results suggest that competing on overdraft prices will not capture market share or increase usage and thus will lower revenue. Although cutting overdraft prices could in principle generate more customer loyalty or reciprocity, the fact that induced overdraft behavior does not persist suggests that these sorts of long-term benefits will not materialize for banks.

More subtly, our results should give pause to third parties seeking to improve overdraft markets with messages (like social marketing campaigns) that draw attention to overdraft costs. To fix ideas, imagine messaging around the theme of “Beware of big overdraft fees!” that is delivered by an entity that might benefit from unshrouding, for example, a regulator, a firm with social objectives or a product-differentiation strategy, or a personal financial management service. Our results are consistent with models of limited and reactive attention where consumer responses to unshrouding are overreactions rather than optimal responses to new information (e.g., Gabaix (2014), Bordalo, Gennaioli, and Shleifer (2015)). Our results further suggest that unshrouding could be quite costly to sustain, since its effects do not persist and incumbent suppliers could effectively counter unshrouding campaigns by advertising nonprice attributes

¹⁸ This is akin to a lower Gabaix and Laibson-type “substitution cost” of avoiding the add-on.

(like availability/credit lines in our case). Accordingly, we are sympathetic to Heidhues, Köszegi, and Murooka's (2017) conjecture that third parties, or deviating firms, will be outgunned in a messaging arms race with incumbent add-on suppliers.

Although our results support policy makers' increasing scrutiny of add-on features, pricing, and practices, we emphasize that we do not conduct the sort of welfare analysis that should motivate and guide policy interventions. One reason we stop short of welfare analysis is that we do not actually have the ability to sharply test existing models of shrouded equilibria, since our experiment considers the existing client base of a single firm rather than competition for customers across firms (Gabaix and Laibson (2006), Armstrong and Vickers (2012), Grubb (2015), Heidhues, Köszegi, and Murooka (2017)).

Future work would do well to focus on welfare, although doing so may require far more household-level consumption and expenditure data than are typically available in studies that rely on administrative data alone. Refining our design could also help test across models, for instance, by testing promotions that mention price without cutting it and by mentioning information on price levels as well as or instead of discounts. It would also be useful to examine consumer perceptions of add-on prices and expectations of usage more directly, given their centrality to theoretical assumptions and predictions.

Appendix: Additional Results

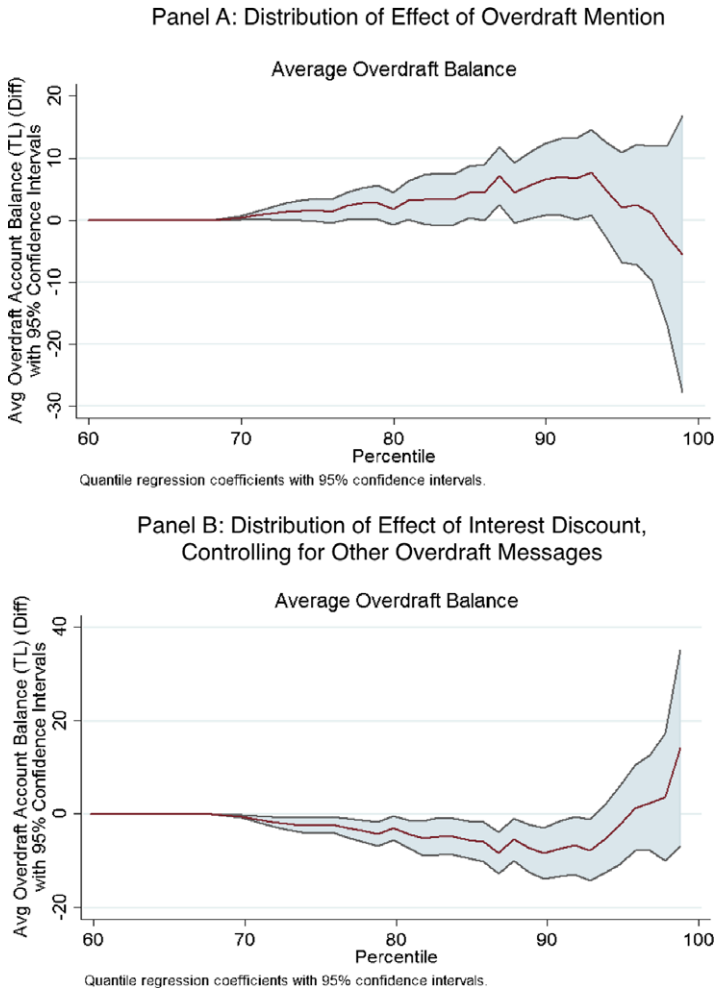


Figure AI. Distribution of effect of overdraft mention and of interest discount. Panel A plots the marginal effects of a message mentioning the overdraft account on each quantile of overdraft account balance. Panel B plots the average marginal effects of an interest discount on each quantile of overdraft account balance. (Color figure can be viewed at wileyonlinelibrary.com)

Table A1

Effects of Marketing on Average Monthly Checking + Savings Balances during Experiment (September to December 2012)

Each column-panel presents treatment effect estimates from a single OLS regression, with Huber–White standard errors (in parentheses), of the average balances measure in the column heading on the experimental variables in the rows. Variables in Panel A rows (4) to (7) identify the additive effect of one or more of the interest discount arms, that is, they identify the difference in treatment effects between the Overdraft Availability message and the Overdraft Interest Discount message. Variables in Panel B rows (2) and (4) identify the additive effect of the overdraft interest discount; that is, they identify the difference in treatment effects between the discount on some other service in row (1) or (3) and that discount bundled with the overdraft interest discount. Balance measures cover September 15 to December 31, 2012. Account balances are in Turkish lira, where 1TL = \$0.56 at the start of the experiment. Figure 1 summarizes the experimental design and shows message scripts. Regressions also include controls for randomization strata. The unit of observation is the unit of randomization: a checking account. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

LHS Variable:	Average Balances		Top 1% Winsorized		Top 1% Dropped		Log	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Overdraft Marketing								
August 30 th Message:								
(1) Overdraft Availability	−8.8782 (9.7697)	−8.8742 (9.7695)	−8.5687* (4.9102)	−8.5665* (4.9100)	−7.3419* (3.7965)	−7.3425* (3.7965)	−0.0118 (0.0115)	−0.0118 (0.0115)
(2) No Message	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted
September 15 Message:								
(3) Overdraft Mention: identifies effect of Overdraft Availability message	−3.5914 (13.9764)	−3.5934 (13.9766)	2.5982 (7.3735)	2.5969 (7.3736)	3.9231 (5.7159)	3.9228 (5.7159)	−0.0082 (0.0171)	−0.0082 (0.0171)
(4) Overdraft Mention, with Interest Discount	11.4448 (12.2665)		3.0329 (6.9651)		1.9943 (5.4186)		0.0108 (0.0162)	
(5) Overdraft Mention, with Interest Discount Only		24.3365 (15.6249)		12.3639 (8.6572)		8.0409 (6.7305)		0.0237 (0.0199)
(6) Overdraft Mention, w/Interest + Auto Bill-Pay Discounts		19.7938 (15.7186)		7.1275 (8.5911)		0.7018 (6.5985)		0.0047 (0.0199)
(7) Overdraft Mention, w/Interest + Debit Card Discounts		−9.7688 (14.9912)		−10.3752 (8.3311)		−2.7430 (6.5513)		0.0042 (0.0197)

(Continued)

Table A1—Continued

LHS Variable:	Average Balances		Top 1% Winsorized		Top 1% Dropped		Log	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Overdraft Marketing								
(8) No Overdraft Mention =	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted
Auto Bill-Pay Discount Only or Debit Card Discount Only								
Mean(LHS)	415,3350	415,3350	359,4495	359,4495	305,0098	305,0098	4,2853	4,2853
SD(LHS)	(1631.25)	(1631.25)	(847.06)	(847.06)	(654.51)	(654.51)	(2.11)	(2.11)
Observations	108,000	108,000	108,000	108,000	106,920	106,920	106,020	106,020
Panel B: Auto Bill-Pay and Debit Card Marketing								
September 15 Message:								
(1) Debit Card Discount (includes both Debit Card Discount Only, and Overdraft Interest Discount + Debit Discount)	−13.0974 (13.5978)		−10.2527 (7.4184)		−12.8017** (5.6403)		−0.0213 (0.0172)	
(2) Debit Card Discount + Overdraft Interest Discount	−8.8192 (15.5592)		−6.2956 (8.3217)		6.0471 (6.4248)		0.0137 (0.0198)	
(3) Auto Bill-Pay Discount (includes both Auto Bill-Pay Discount Only, and Overdraft Interest Discount + Auto Bill-Pay Discount)	−4.0234 (17.0081)		−7.2920 (7.3615)		−3.0861 (5.7486)		0.0140 (0.0172)	
(4) Auto Bill-Pay Discount + Overdraft Interest Discount	11.6737 (19.2085)		8.2507 (8.5362)		−0.2214 (6.5709)		−0.0211 (0.0199)	
(5) Overdraft Interest Discount Only	Omitted		Omitted		Omitted		Omitted	
Controls for Aug 30 and Sep 15 Overdraft Availability	yes		yes		yes		yes	
Only Messages								
Mean(LHS)	415,3350		359,4495		305,0098		4,2853	
SD(LHS)	(1631.25)		(847.06)		(654.51)		(2.11)	
Observations	108,000		108,000		106,920		106,020	

Table A2

Effects of Overdraft Marketing on Number of Active Products during Experiment (September to December 2012)

Each column presents treatment effect estimates from a single OLS regression, with Huber–White standard errors (in parentheses), of the measure of the account holder's active products at the experimenting bank (including the checking account that is our unit of observation) in the column heading on the experimental variables in the rows. Variables in rows (4) to (7) identify the additive effect of one or more of the interest discount arms, that is, they identify the difference in treatment effects between the Overdraft Availability message and the Overdraft Interest Discount message. Figure 1 summarizes the experimental design and shows message scripts. Regressions also include controls for randomization strata. The unit of observation is the unit of randomization: a checking account. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

LHS Variable:	Count		Log	
	(1)	(2)	(3)	(4)
August 30th Message:				
(1) Overdraft Availability	0.0131 (0.0136)	0.0131 (0.0136)	0.0028 (0.0047)	0.0028 (0.0047)
(2) No Message	Omitted	Omitted	Omitted	Omitted
September 15 Message:				
(3) Overdraft Mention: identifies effect of Overdraft Availability message	−0.0002 (0.0204)	−0.0002 (0.0204)	−0.0005 (0.0070)	−0.0005 (0.0070)
(4) Overdraft Mention, with Interest Discount	0.0155 (0.0192)		0.0093 (0.0066)	
(5) Overdraft Mention, with Interest Discount only		0.0145 (0.0235)		0.0074 (0.0081)
(6) Overdraft Mention, w/Interest + Auto Bill-Pay Discounts		−0.0073 (0.0234)		0.0027 (0.0081)
(7) Overdraft Mention, w/Interest + Debit Card Discounts		0.0393* (0.0236)		0.0177** (0.0081)
(8) No Overdraft Mention =	Omitted	Omitted	Omitted	Omitted
Auto Bill-Pay Discount Only or Debit Card Discount Only				
Mean(LHS)	2.6238	2.6238	0.6348	0.6348
SD(LHS)	(2.27)	(2.27)	(0.78)	(0.78)
Observations	108,000	108,000	108,000	108,000

Table A3
Does the August 30 Message Mediate Treatment Effects of Later Messages?

Each column presents treatment effect estimates from a single OLS regression, with Huber–White standard errors (in parentheses), of the usage measure in the column heading on the experimental variables in the rows. Variables in rows (4) to (11) identify the additive effect of one or more of the interest discount arms, this is, they identify the difference in treatment effects between the Overdraft Availability message and the Overdraft Interest Discount. Overdraft usage measures cover September 15 to December 31, 2012. Account balances are in Turkish lira, where 1TL = \$0.56 at the start of the experiment. Figure 1 summarizes the experimental design and shows message scripts. Regressions also include controls for randomization strata. The unit of observation is the unit of randomization: a checking account. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

LHS Variable: Measured during Experiment: Over September 15 to December 31, 2012	Overdraft Account Used		Days with Overdraft Balance		Mean(Overdraft Account Balance)	
	(1)	(2)	(3)	(4)	(5)	(6)
August 30 th Message: (1) Overdraft Availability	−0.0001 (0.0049)	−0.0001 (0.0049)	0.0477 (0.0631)	0.0476 (0.0631)	1.1445 (0.7875)	1.1444 (0.7875)
No Message	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted
September 15 Message: (2) Overdraft Mention with Aug 30 Msg	0.0114* (0.0060)	0.0114* (0.0060)	0.2021** (0.0797)	0.2021** (0.0797)	0.7911 (1.0046)	0.7912 (1.0046)
(3) Overdraft Mention without Aug 30 Msg	0.0063 (0.0060)	0.0063 (0.0060)	0.0622 (0.0767)	0.0622 (0.0767)	1.1209 (0.9814)	1.1212 (0.9815)
(4) Overdraft Mention, with Interest Discount, with Aug 30 Msg	−0.0151*** (0.0056)	−0.0151*** (0.0056)	−0.2149*** (0.0755)	−0.2149*** (0.0755)	−0.6116 (0.9576)	−0.6116 (0.9576)
(5) Overdraft Mention, with Interest Discount, w/o Aug 30 Msg	−0.0098* (0.0056)	−0.0098* (0.0056)	−0.1067 (0.0722)	−0.1067 (0.0722)	−0.8735 (0.9420)	−0.8735 (0.9420)
(6) Overdraft Interest Discount, No Other Discount, with Aug 30 Msg		−0.0082 (0.0069)		−0.1819** (0.0916)		−0.9928 (1.1290)
(7) Overdraft Interest Discount, No Other Discount, w/o Aug 30 Msg		−0.0056 (0.0069)		0.0109 (0.0895)		0.3160 (1.1630)
(8) Overdraft Interest Discount + Auto Bill-Pay Discount, w/Aug 30 Msg		−0.0135* (0.0069)		−0.1894** (0.0922)		0.0209 (1.1848)

(Continued)

Table A3—Continued

LHS Variable: Measured during Experiment: Over September 15 to December 31, 2012	Overdraft Account Used		Days with Overdraft Balance		Mean(Overdraft Account Balance)	
	(1)	(2)	(3)	(4)	(5)	(6)
(9) Overdraft Interest Discount + Auto Bill-Pay Discount, w/o Aug 30 Msg		−0.0139** (0.0069)		−0.1875** (0.0875)		−2.2005* (1.1330)
(10) Overdraft Interest Discount + Debit Card Discount, with Aug 30 Msg		−0.0236*** (0.0069)		−0.2734*** (0.0911)		−0.8622 (1.2162)
(11) Overdraft Interest Discount + Debit Card Discount, w/o Aug 30 Msg		−0.0098 (0.0069)		−0.1434 (0.0877)		−0.7351 (1.1541)
No Overdraft Mention = Auto Bill-Pay Discount Only or Debit Card Discount Only	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted
<i>p</i> -Value on <i>F</i> -test of equality between rows (2) and (3)	0.5475	0.5477	0.2058	0.2058	0.8142	0.8140
<i>p</i> -Value on <i>F</i> -test of equality between rows (4) and (5)	0.5031		0.2999		0.8454	
<i>p</i> -Value on <i>F</i> -test of equality between rows (6) and (7)		0.7870		0.1323		0.4192
<i>p</i> -Value on <i>F</i> -test of equality between rows (8) and (9)		0.9644		0.9875		0.1755
<i>p</i> -Value on <i>F</i> -test of equality between rows (10) and (11)		0.1556		0.3038		0.9395
Mean(LHS)	0.3077	0.3077	2.7676	2.7676	26.8511	26.8511
SD(LHS)	(0.46)	(0.46)	(6.05)	(6.05)	(77.97)	(77.97)
Observations	108,000	108,000	108,000	108,000	108,000	108,000

Table A4
**Effects of Overdraft Marketing Frequency on Overdraft Usage During Experiment
(September to December 2012)**

Each column presents treatment effect estimates from a single OLS regression, with Huber–White standard errors (in parentheses), of the usage measure in the column heading on the experimental variables in the rows. Variables in rows (5) to (8) identify the additive effect of one or more of the interest discount arms, this is, they identify the difference in treatment effects between the Overdraft Availability message and the Overdraft Interest Discount message. Usage measures cover September 15 to December 31, 2012. Account balances are in Turkish lira, where 1TL = \$0.56 at the start of the experiment. Figure 1 summarizes the experimental design and shows message scripts. Regressions also include controls for randomization strata. The unit of observation is the unit of randomization: a checking account. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

LHS Variable:	Overdraft Account Used		Days with Overdraft Balance		Mean(Overdraft Account Balance)	
	(1)	(2)	(3)	(4)	(5)	(6)
September 15 Message:						
(1) Overdraft Mention No Subsequent Messages	0.0003 (0.0064)	0.0003 (0.0064)	0.0429 (0.0835)	0.0429 (0.0835)	-0.3108 (1.0420)	-0.3108 (1.0420)
(2) Overdraft Mention + Subsequent Messages	0.0132*** (0.0049)	0.0132*** (0.0049)	0.1765*** (0.0644)	0.1765*** (0.0644)	1.5872* (0.8283)	1.5872* (0.8283)
(3) Overdraft Mention + Frequent Messages		0.0136** (0.0065)		0.2119** (0.0854)		1.4980 (1.0996)
(4) Overdraft Mention + Infrequent Messages		0.0127** (0.0065)		0.1411* (0.0855)		1.6763 (1.1017)
(5) Overdraft Mention, with Interest Discount, No Subsequent Messages	-0.0017 (0.0069)	-0.0017 (0.0069)	0.0386 (0.0898)	0.0385 (0.0898)	1.8416 (1.1400)	1.8416 (1.1400)
(6) Overdraft Mention, with Interest Discount + Subsequent Messages	-0.0178*** (0.0049)	-0.0178*** (0.0049)	-0.2602*** (0.0642)	-0.2602*** (0.0642)	-2.0319** (0.8315)	-2.0319** (0.8315)
(7) Overdraft Mention, with Interest Discount + Frequent Messages		-0.0207*** (0.0069)		-0.3368*** (0.0906)		-2.0617* (1.1763)
(8) Overdraft Mention, with Interest Discount + Infrequent Messages		-0.0149** (0.0069)		-0.1836** (0.0909)		-2.0021* (1.1706)
(9) No Overdraft Mention = Auto Bill-Pay Discount Only or Debit Card Discount Only	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted
Mean(LHS)	0.3077 (0.46)	0.3077 (0.46)	2.7676 (6.05)	2.7676 (6.05)	26.8511 (77.97)	26.8511 (77.97)
SD(LHS)	108,000	108,000	108,000	108,000	108,000	108,000
Observations						

Table A5

Effects of Overdraft Message and Discount Duration on Overdraft Usage during Experiment
(September to December 2012)

Each column presents treatment effect estimates from a single OLS regression, with Huber–White standard errors, of the usage measure in the column heading on the experimental variables in the rows. Variables in rows (5) to (10) identify the additive effect of one or more of the interest discount arms; that is, they identify the difference in treatment effects between the Overdraft Availability message and the Overdraft Interest Discount message. Usage measures cover September 15 to December 31, 2012. Account balances are in Turkish lira, where 1TL = \$0.56 at the start of the experiment. Figure 1 summarizes the experimental design and shows message scripts. Regressions also include controls for randomization strata. The unit of observation is the unit of randomization: a checking account. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

LHS Variable:	Overdraft Account Used		Days with Overdraft Balance		Mean(Overdraft Account Balance)	
	(1)	(2)	(3)	(4)	(5)	(6)
September 15 Message:						
(1) Overdraft Mention: identifies effect of Overdraft Availability Message	0.0089** (0.0042)		0.1320** (0.0553)		0.9556 (0.7029)	
(2) Overdraft Mention, long duration, with Subsequent Messages		0.0154** (0.0065)		0.1718** (0.0857)		3.1076*** (1.1831)
(3) Overdraft Mention, short duration, with Subsequent Messages		0.0109* (0.0065)		0.1811** (0.0853)		0.0711 (1.0163)
(4) Overdraft Mention No Subsequent Messages		0.0003 (0.0064)		0.0429 (0.0835)		−0.3108 (1.0420)
(5) Overdraft Mention, with Interest Discount, long duration	−0.0134*** (0.0044)		−0.1467** (0.0583)		−1.0662 (0.7510)	
(6) Overdraft Mention, with Interest Discount, short duration	−0.0114** (0.0044)		−0.1747*** (0.0580)		−0.4189 (0.7516)	
(7) Overdraft Mention, with Interest Discount, long duration, with Subsequent Messages		−0.0227*** (0.0069)		−0.2671*** (0.0910)		−4.2866*** (1.2475)
(8) Overdraft Mention, with Interest Discount, short duration, with Subsequent Messages		−0.0129* (0.0069)		−0.2533*** (0.0905)		0.2155 (1.0995)

(Continued)

Table A5—Continued

LHS Variable:	Overdraft Account Used		Days with Overdraft Balance		Mean(Overdraft Account Balance)	
	(1)	(2)	(3)	(4)	(5)	(6)
(9) Overdraft Mention, with Interest Discount, long duration, w/o Subsequent Messages		0.0006 (0.0077)		0.1033 (0.1014)		2.3327* (1.3053)
(10) Overdraft Mention, with Interest Discount, short duration, w/o Subsequent Messages		-0.0040 (0.0077)		-0.0263 (0.1001)		1.3480 (1.2777)
(11) No Overdraft Mention = Auto Bill-Pay Discount Only or Debit Card Discount Only	Omitted	Omitted	Omitted	Omitted	Omitted	Omitted
<i>p</i> -Value on <i>F</i> -test of equality between rows (2) and (3)		0.5931		0.9344		0.0371
<i>p</i> -Value on <i>F</i> -test of equality between rows (2) and (4)		0.0738		0.2454		0.0206
<i>p</i> -Value on <i>F</i> -test of equality between rows (3) and (4)		0.2094		0.2118		0.7764
<i>p</i> -Value on <i>F</i> -test of equality between rows (5) and (6)	0.6183		0.5858		0.3359	
<i>p</i> -Value on <i>F</i> -test of equality between rows (7) and (8)		0.3159		0.9141		0.0068
<i>p</i> -Value on <i>F</i> -test of equality between rows (7) and (9)		0.0244		0.0065		0.0002
<i>p</i> -Value on <i>F</i> -test of equality between rows (7) and (10)		0.0711		0.0750		0.0016
<i>p</i> -Value on <i>F</i> -test of equality between rows (8) and (9)		0.1919		0.0087		0.2145
<i>p</i> -Value on <i>F</i> -test of equality between rows (8) and (10)		0.3910		0.0926		0.5018
<i>p</i> -Value on <i>F</i> -test of equality between rows (9) and (10)		0.5010		0.1555		0.4174
Observations	108,000	108,000	108,000	108,000	108,000	108,000

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Data and Scripts.